Financial Planner — Architecture

Services, data flows, shipping workflow, and data model

Production: planner.boombaleia.com · Source of truth: `docs/architecture.md`

Status badges used throughout this document

ACTIVE running today in production at `planner.boombaleia.com`.
WIRED-DORMANT code and configuration are shipped, but the service is idle until a credential (DSN / API key / secret) is added to Vercel. Turning it on is a one-variable change, no redeploy needed if configured at the project level.
PLANNED not yet built. The M1 / M2 / M3 / M4 tag indicates the milestone that introduces it.

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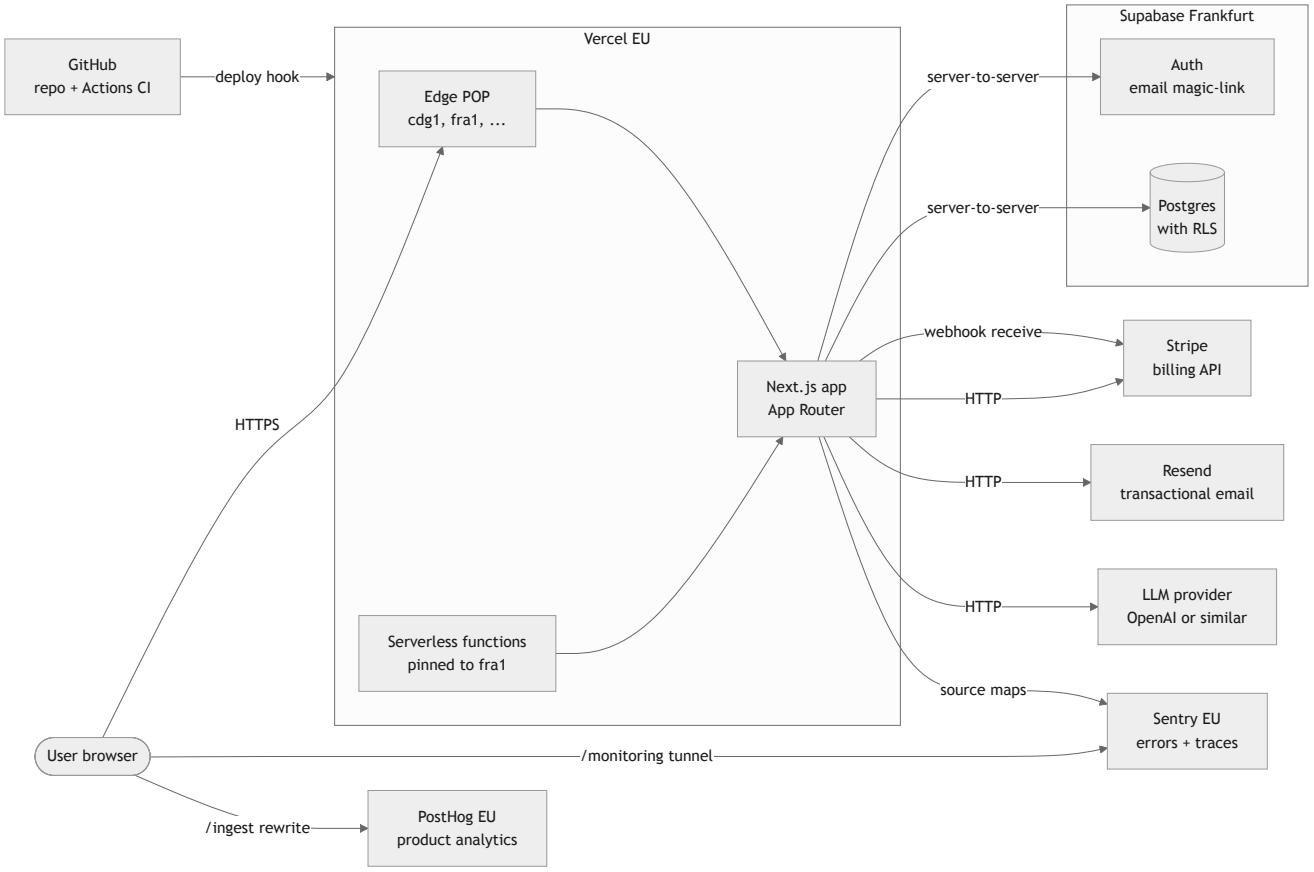
System overview

Financial Planner is a single-tenant-per-user web application: each signed-in user owns a set of financial scenarios, runs deterministic projections plus probabilistic Monte Carlo simulations on them, and (from M3) can query an AI assistant scoped to their own numbers.

The stack is intentionally small and boring: **Next.js 15 App Router on Vercel** as the runtime; **Supabase Postgres** as the database with **Row-Level Security (RLS)** enforcing tenant isolation in

the database itself; **Stripe** for billing; **Sentry + PostHog** for observability. Everything that stores user data runs in the EU (Frankfurt) for GDPR simplicity.

Context diagram



The shaded `Vercel EU` and `Supabase Frankfurt` groups are the two stateful / compute-heavy substrates; everything else is a stateless third-party used from Next.js server code or directly from the browser via a first-party proxy path.

Services and what each one captures

Service	Purpose	Region	Data captured	Status
Vercel	Hosting + CDN + serverless runtime	Edge global, functions `fra1`	Access logs (IP, URL, status), build logs	ACTIVE
Next.js app	UI, server actions, API routes	Runs inside Vercel	Request lifecycle only (nothing persisted by default)	ACTIVE
`@app/core` package	Pure domain logic (schema + projection math)	N/A (library)	Nothing; it's pure functions	ACTIVE
Supabase Auth	User sign-in (email magic-link)	Frankfurt	Email, hashed password (if used), session rows	PLANNED (M1)
Supabase Postgres	Scenarios, simulations, subscriptions, chat history	Frankfurt	All durable user content	PLANNED (M1)
Stripe	Subscription billing	US (PCI-scoped)	Customer record, subscription state, payment methods	PLANNED (M4)
Resend	Transactional email	EU	Email delivery logs (from, to, subject, status)	PLANNED (M1)
Sentry	Error + performance telemetry	EU	Exception stack traces, perf spans, release metadata	WIRED-DORMANT
PostHog	Product analytics	EU Cloud	`$pageview`, `$pageleave`, explicit `capture(...)` calls	WIRED-DORMANT
LLM provider	AI assistant	US most likely	User prompts, scenario numbers, assistant replies	PLANNED (M3)
GitHub	Source of truth + CI	Global	Source code, CI logs, PR metadata	ACTIVE
Gandi.net	DNS authoritative for `boombaleia.com`	Global	Only `planner` CNAME → Vercel	ACTIVE

2.1 Vercel ACTIVE

Hosts the Next.js app. Static assets are served from the global edge network (whichever POP is closest to the user — verified `cdg1` for requests from France). Serverless functions are pinned to `fra1` via `vercel.json` so they execute geographically next to Supabase:

```
{ "regions": ["fra1"], "framework": "nextjs", "installCommand": "npm install" }
```

Vercel auto-deploys `main` to production and every pull request to a preview URL. It also provisions and renews the TLS certificate for `planner.boombaleia.com`.

What it sees: access logs (IP, URL, status, UA, response time), function invocation logs, build logs. No user data goes through Vercel storage — it's a pure pass-through.

2.2 Next.js app ACTIVE

App Router layout, Client + Server Components, server actions, and a handful of API routes:

- `app/src/app/page.tsx` — landing + planner UI
- `app/src/app/api/billing/checkout/route.ts` — creates a Stripe Checkout session
- `app/src/app/api/webhooks/stripe/route.ts` — receives Stripe webhook events
- `app/src/app/global-error.tsx` — top-level error boundary that reports to Sentry
- `app/src/instrumentation.ts` — Next.js instrumentation hook initializing Sentry on boot

Feature code is organised under `app/src/features/{planner,currency,auth,billing,projects,settings}/` so each concern has its own folder and tests.

2.3 `@app/core` package ACTIVE

A separate npm workspace at `packages/core/` that owns the domain model and financial math. It has no React, no environment access, no I/O — making it trivially unit-testable (131/131 passing in CI) and reusable across:

- the client bundle (forms and charts)
- server actions (validation before persistence)
- future Supabase Edge Functions (if we ever run projections server-side at scale)
- the AI assistant's tool layer (`runProjection` as an LLM tool)

Exports: `PlanInputsSchema`, `PlanInputs`, `ProjectionPoint`, `DEFAULT_PLAN_INPUTS`, `projectNetWorth`, `deflateToToday`, `ageFromDob`, `clampHorizon` (see `packages/core/src/index.ts`).

2.4 Supabase Auth PLANNED (M1)

Hosted Postgres project in Frankfurt (region `eu-central-1`) with the built-in **GoTrue** auth service. Plan: email magic-link as the primary sign-in method; OAuth (Google) as a stretch goal.

What it will capture: email address, hashed password if one is set, per-device refresh tokens, audit log (sign-in timestamps + IPs) that Supabase retains for 90 days by default.

Session is transported via an HttpOnly cookie set by `@supabase/ssr` so both Server Components and Route Handlers can read it. No long-lived JWT ever enters `localStorage`.

2.5 Supabase Postgres PLANNED (M1)

Same Supabase project. **Row-Level Security (RLS) is the primary tenant-isolation mechanism** — every table has an `owner_id uuid references auth.users(id)` column and a policy of the form:

```
alter table scenarios enable row level security;
create policy "users read own scenarios"  on scenarios for select using (auth.uid() =
create policy "users write own scenarios" on scenarios for all    using (auth.uid() =
```

With RLS enabled, even a bug that accidentally queries `select * from scenarios` without a `where` clause is safe: Postgres silently filters to the current user's rows.

See the full planned schema in Data model.

2.6 Stripe PLANNED (M4)

Checkout-hosted flow (no PCI scope on our side). The checkout route is already scaffolded at `app/src/app/api/billing/checkout/route.ts`. Webhook signatures are verified with `STRIPE_WEBHOOK_SECRET` in `app/src/app/api/webhooks/stripe/route.ts`.

What it will capture: customer record (id, email), subscription state, payment method metadata, invoices. We never see raw card data.

2.7 Resend PLANNED (M1)

Transactional email: welcome / magic-link / password-reset (if we offer a password path) / billing receipts. EU region. DKIM + DMARC records live on Gandi alongside the `planner` CNAME.

What it will capture: message metadata only (from, to, subject, delivery status). Content of the emails is rendered from templates in `app/src/features/auth/emails/` and kept deliberately short to minimise PII exposure.

2.8 Sentry WIRED-DORMANT

Error + performance telemetry, EU region (`ingest.de.sentry.io`). The SDK is fully wired but dormant until `NEXT_PUBLIC_SENTRY_DSN` is set in Vercel — see `app/src/instrumentation.ts`

(server + edge), `app/src/instrumentation-client.ts` (browser), and the `withSentryConfig` wrapper in `app/next.config.ts`.

Browser → Sentry traffic goes through a first-party **tunnel route** at `/monitoring` so ad blockers don't silently drop events (many block direct requests to `*.sentry.io`). The route is auto-generated by Sentry's Next.js plugin at build time when a DSN is present.

What it will capture when enabled: exception stack traces (with source maps), 10% of request traces for perf baselines, release + deployment metadata, breadcrumbs (URL changes, fetch calls). No form values, no email addresses — Sentry's PII scrubbing is on by default and we never call `Sentry.setUser({ email })`.

2.9 PostHog WIRED-DORMANT

Product analytics, EU Cloud (`eu.i.posthog.com`). The provider is mounted in `app/src/components/PostHogProvider.tsx` with three deliberate privacy choices:

1. **`persistence: "memory"`** — no cookies, no localStorage. A user who refreshes gets a new anonymous id. Trade-off: no cross-session funnel analysis, but also no consent banner required on day one. M1 will switch to `"localStorage"` behind a proper consent CMP.
2. **`autocapture: false`** — every event is explicit. Easier to reason about what leaves the browser.
3. **`capture_pageview: false`** + manual pageview tracking — the App Router's client-side navigations don't trigger real page loads, so we capture `$pageview` manually in `PageViewTracker` on pathname/search changes.

Browser → PostHog traffic goes through a first-party rewrite at `/ingest/*` (configured in `app/next.config.ts`) for the same ad-blocker reason as Sentry.

What it will capture when enabled: `$pageview`, `$pageleave`, and explicit `capture(event, props)` calls we make from feature code. Never user identities; never form input values.

2.10 LLM provider PLANNED (M3)

OpenAI (or equivalent) behind a server-only route. The prompt will include the user's current scenario numbers as JSON (no PII like name/email). Function-calling tools are implemented in `@app/core` so the model can invoke `runProjection` and get structured results back.

2.11 GitHub + GitHub Actions ACTIVE

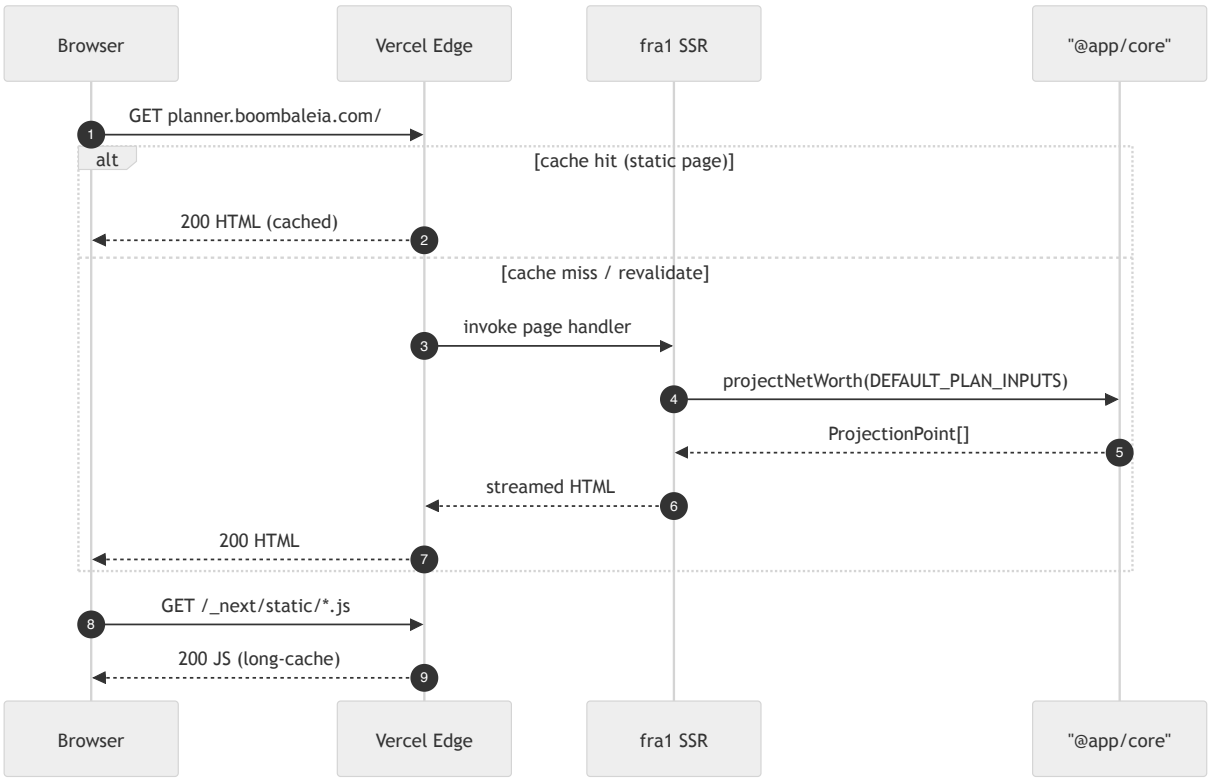
Source of truth at `github.com/lejeff/financial-planner`. Branch protection on `main` requires a PR, 1 approval, linear history, and green status checks. CI is defined in `.github/workflows/ci.yml` and runs two jobs:

1. `checks` — lint, typecheck, unit tests, build (~2 min)
2. `e2e` — Playwright suite against a real built Next.js server (~1.5 min)

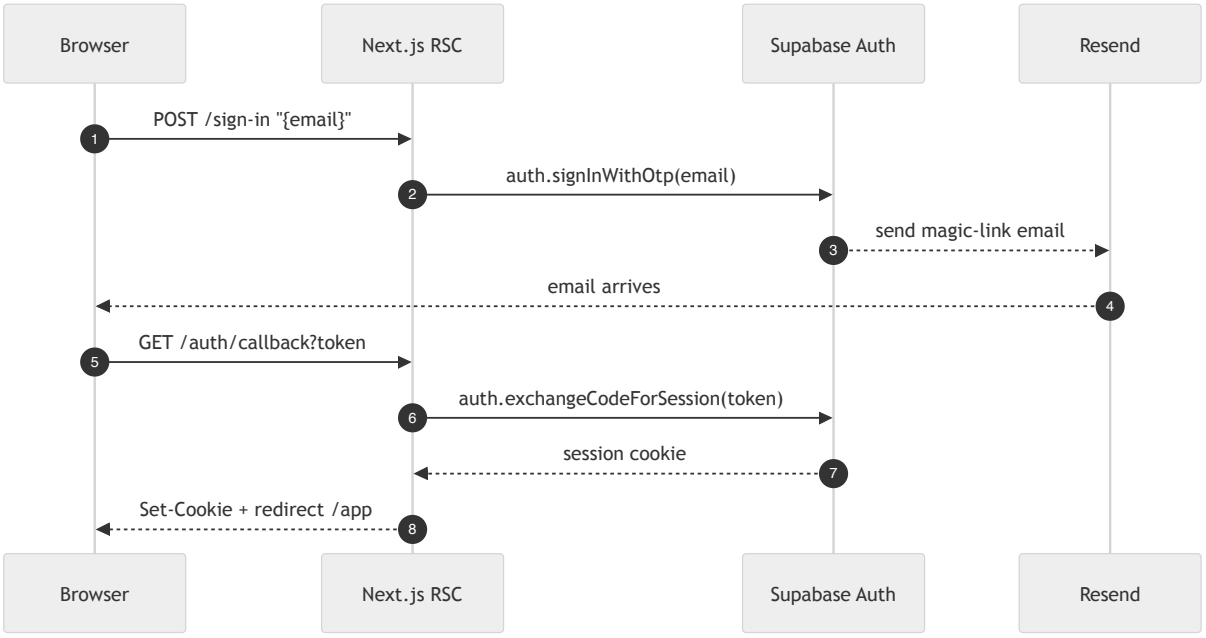
The CI `env:` block carries well-formed placeholders for the strict env schema so a build can exercise the full prod path without needing real secrets.

Data flows (sequence diagrams)

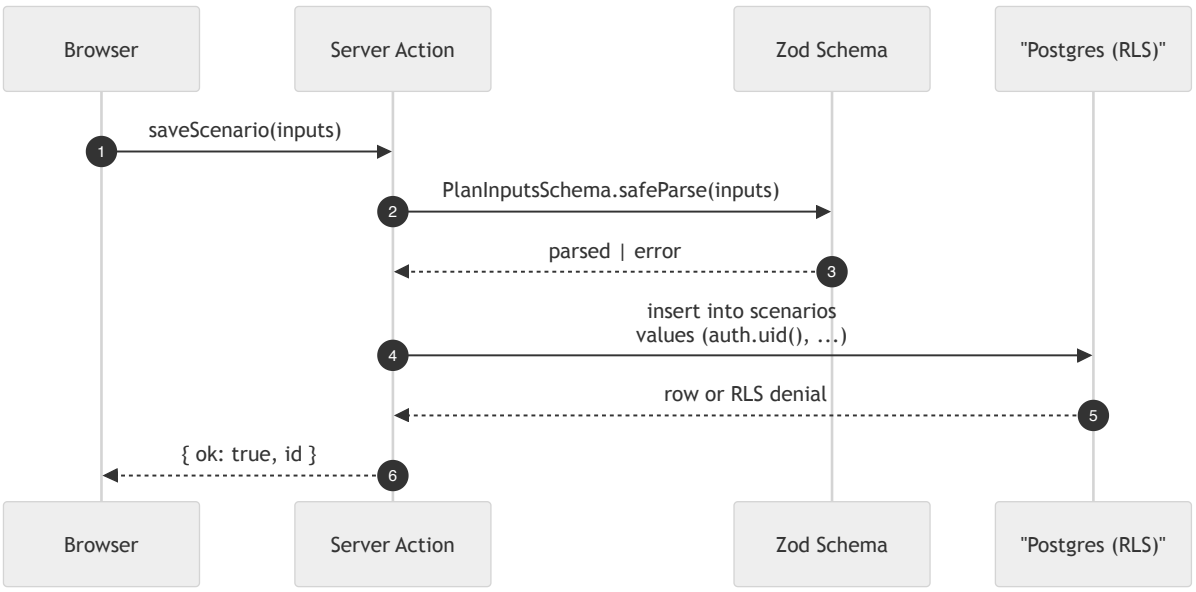
3.1 Cold page load ACTIVE



3.2 Sign-in flow PLANNED (M1)

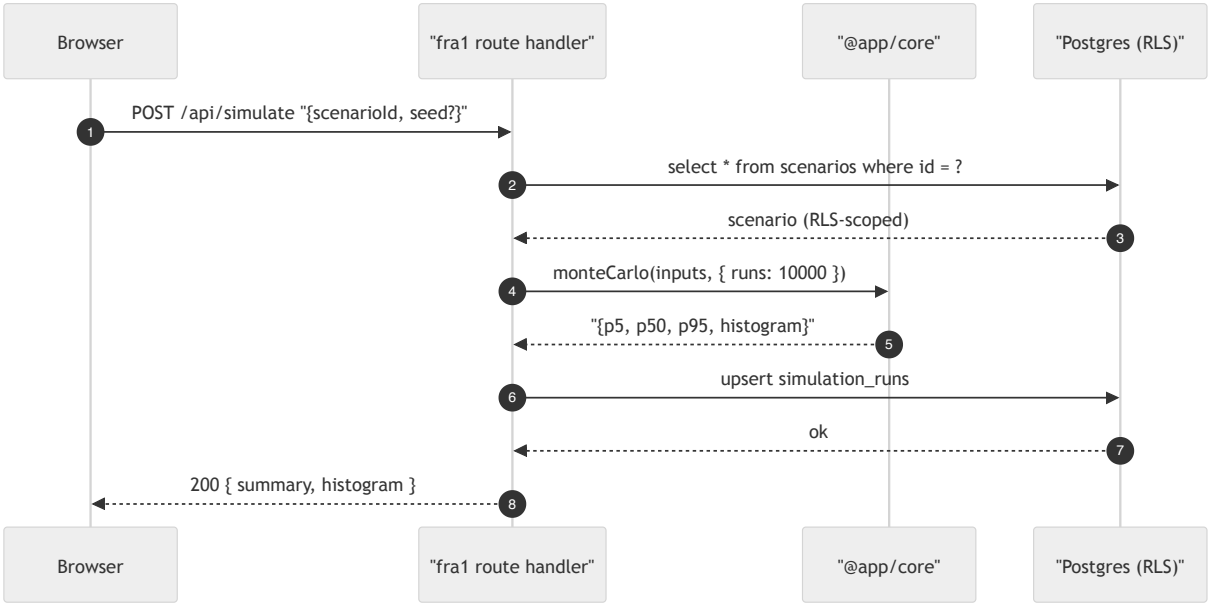


3.3 Save a scenario PLANNED (M1)

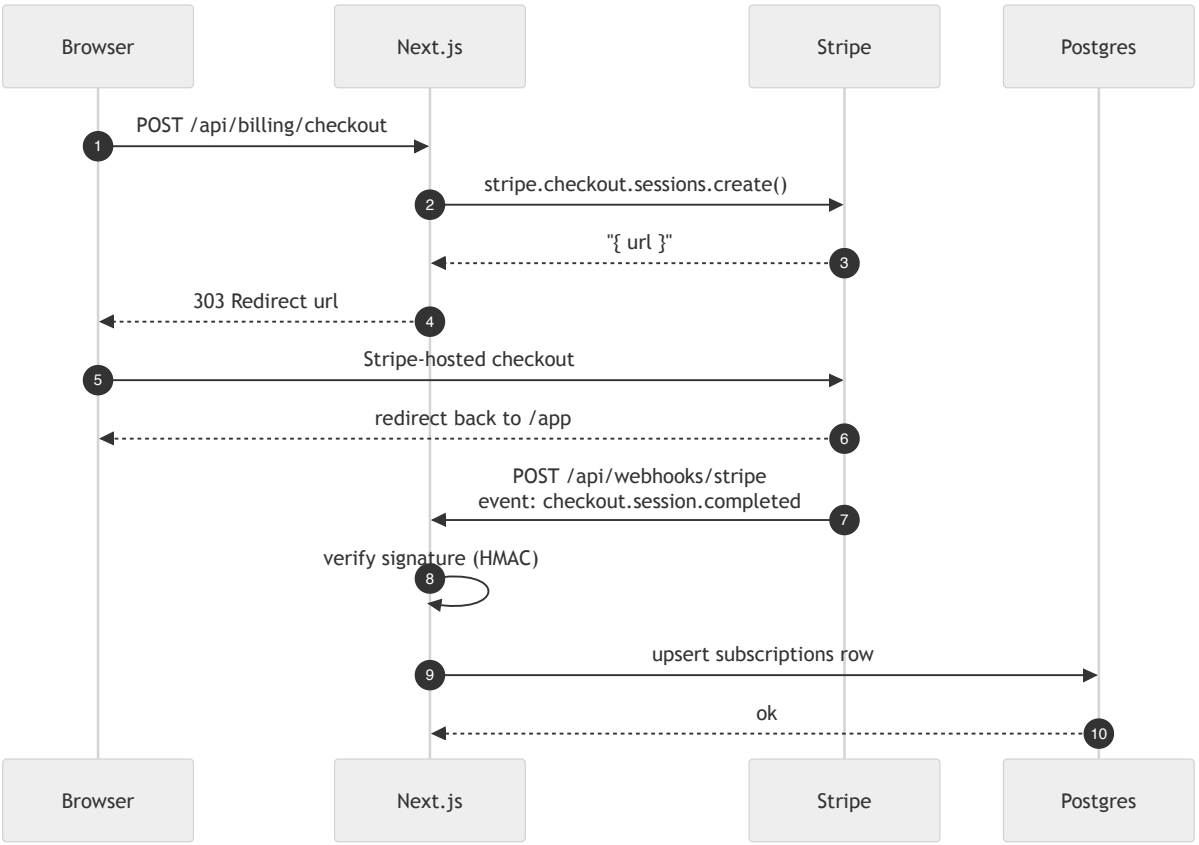


Note: the `auth.uid()` in the insert is enforced by a Postgres `check (owner_id = auth.uid())` constraint, not supplied by the client. A compromised client cannot forge ownership.

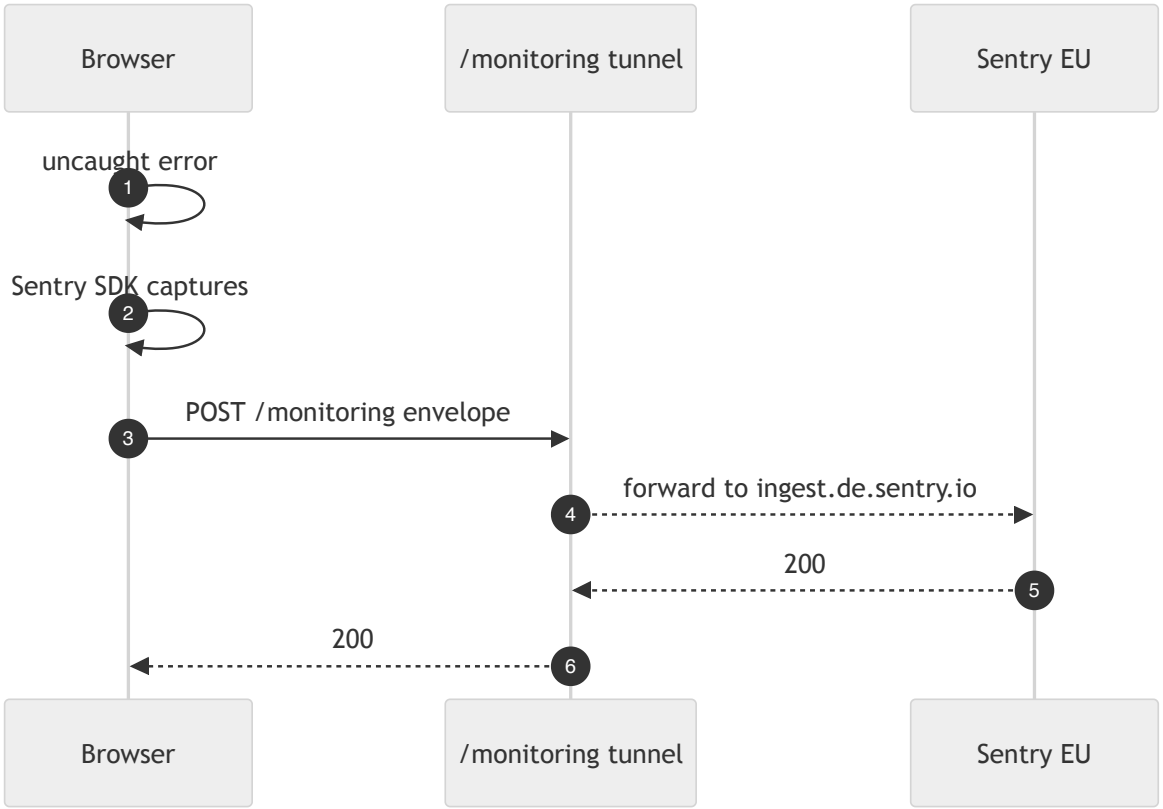
3.4 Run a Monte Carlo simulation PLANNED (M2)



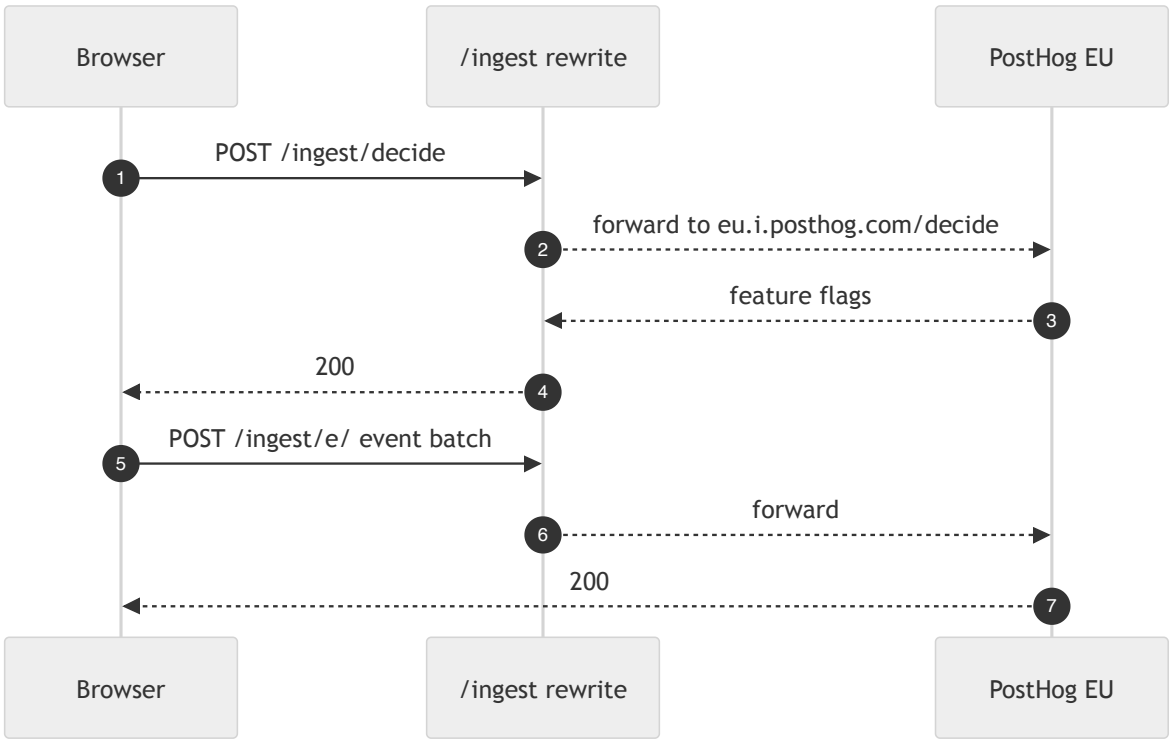
3.5 Stripe checkout + webhook PLANNED (M4)



3.6 Client error → Sentry WIRED-DORMANT



3.7 Client event → PostHog WIRED-DORMANT



Data model

4.1 Today ACTIVE

The only persistent data model today is `PlanInputs`, defined as a Zod schema in `packages/core/src/planInputs.ts` and persisted to the browser's localStorage by `app/src/features/planner/storage.ts`.

Field	Type	Units	Constraints
`name`	string	—	any
`dateOfBirth`	string	YYYY-MM-DD	regex-checked
`startAssets`	number	currency	≥ 0
`startDebt`	number	currency	≥ 0
`debtInterestRate`	number	fraction	0 … 0.20
`debtRepaymentType`	enum	—	`"overTime"` | `"inFine"`
`debtEndYear`	number	year	integer
`monthlySpending`	number	currency	≥ 0
`annualIncome`	number	currency	≥ 0
`retirementAge`	number	years	18 … 100, integer
`rentalIncome`	number	currency	≥ 0
`rentalIncomeRate`	number	fraction	−0.05 … 0.10
`nominalReturn`	number	fraction	−0.5 … 0.5
`inflationRate`	number	fraction	−0.05 … 0.15
`horizonYears`	number	years	10 … 80
`cashBalance`	number	currency	≥ 0
`nonLiquidInvestments`	number	currency	≥ 0
`otherFixedAssets`	number	currency	≥ 0
`nonLiquidLiquidityYear`	number	year	integer
`otherFixedLiquidityYear`	number	year	integer
`realEstateHoldings`	`RealEstateHolding[]`	—	stackable list of currently-owned properties (see below); default `[]`
`events`	`LifeEvent[]`	—	discriminated union (see below); default `[]`

`REALESTATEHOLDING`

A property the user owns today: a `value` (in today's money) plus an annual `appreciationRate`. Each entry compounds independently from year 0; the engine sums them

into the same `realEstate` bucket as any `RealEstateInvestmentEvent` contributions. No purchase deduction (the asset is already owned) and no rental flow (rental belongs on `RealEstateInvestmentEvent`). Each carries a stable `id` (uuid) so the form can edit/remove a specific entry across renders.

Field	Type	Units	Constraints
`id`	string	—	non-empty (uuid in practice)
`type`	literal	—	`"realEstateHolding"`
`value`	number	currency	≥ 0
`appreciationRate`	number	fraction	−0.05 … 0.10

`LIFEEVENT` VARIANTS

A discriminated union keyed by `type`, persisted as part of `PlanInputs`. Each variant carries a stable `id` (uuid) so the form can edit/remove a specific entry across renders. There are two variants today; new variants drop in by adding another schema to `LifeEventSchema` in `packages/core/src/planInputs.ts`.

`RealEstateInvestmentEvent` (`type: "realEstateInvestment"`)

A future property purchase: at `purchaseYear` the engine deducts `purchaseAmount` (in today's money, inflated to the landing year) from the liquid portfolio, seeds a per-event property bucket and rental stream, then compounds both at the per-event rates each subsequent year. Behaves like a held property + `rentalIncome` but starts in the future and is funded from liquid assets.

Field	Type	Units	Constraints
`id`	string	—	non-empty (uuid in practice)
`type`	literal	—	`"realEstateInvestment"`
`purchaseAmount`	number	currency (today's money)	≥ 0
`purchaseYear`	number	year	integer
`appreciationRate`	number	fraction	−0.05 … 0.10
`annualRentalIncome`	number	currency (today's money)	≥ 0
`rentalIncomeRate`	number	fraction	−0.05 … 0.10

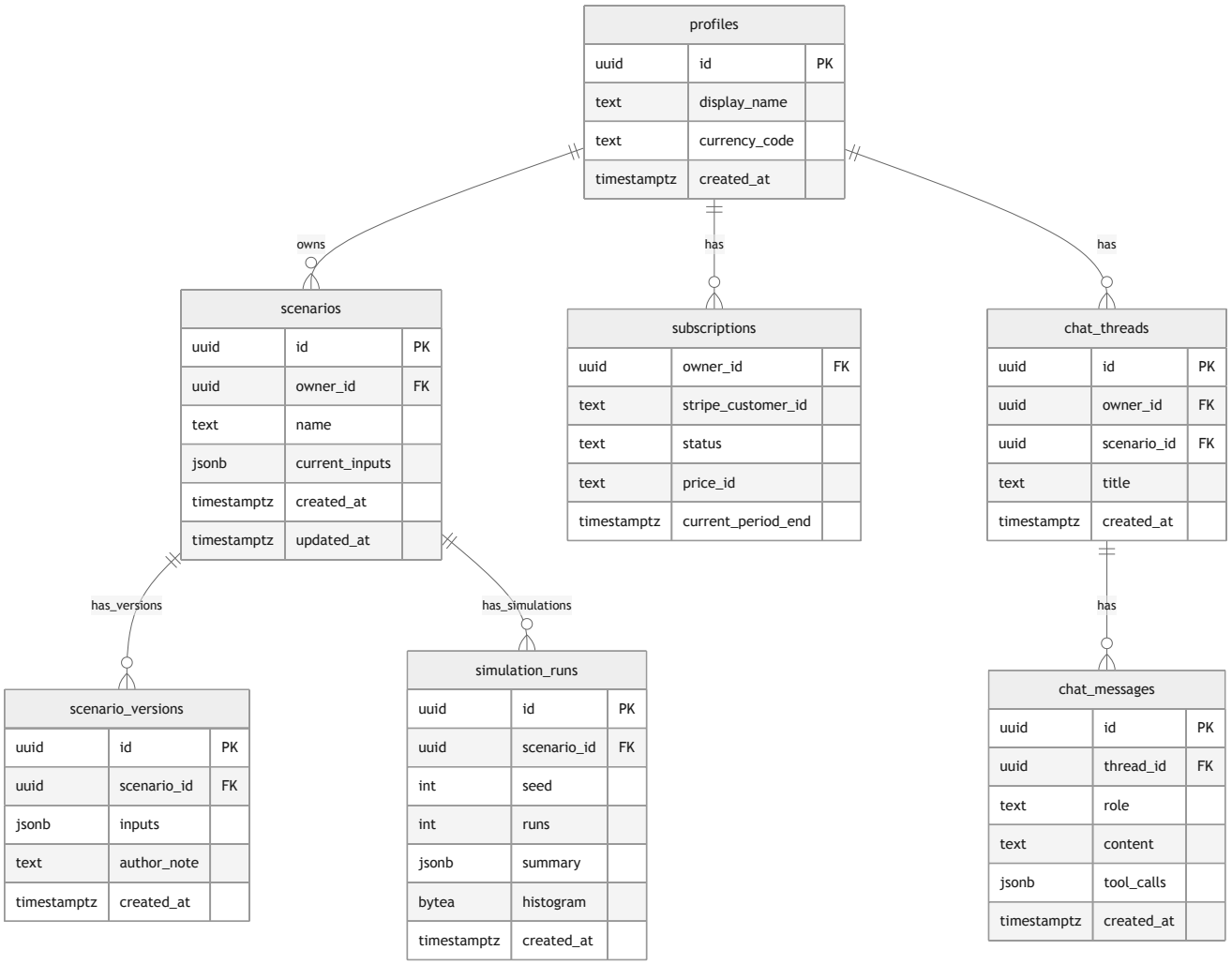
`WindfallEvent` (`type: "windfall"`)

A one-off cash deposit landing in the liquid portfolio at year-end of `year`. The amount is entered in today's money and the engine inflates it to the landing year (same convention as `RealEstateInvestmentEvent.purchaseAmount`). Multiple windfall events stack independently; each fires once when its `year` matches the projection's calendar year.

Field	Type	Units	Constraints
`id`	string	—	non-empty (uuid in practice)
`type`	literal	—	`"windfall"`
`amount`	number	currency (today's money)	≥ 0
`year`	number	year	integer

Derived output: `ProjectionPoint[]` with `{ year, age, netWorth, liquid, savings, otherAssets, realEstate, debt }`.

4.2 Target (M1 → M4) PLANNED



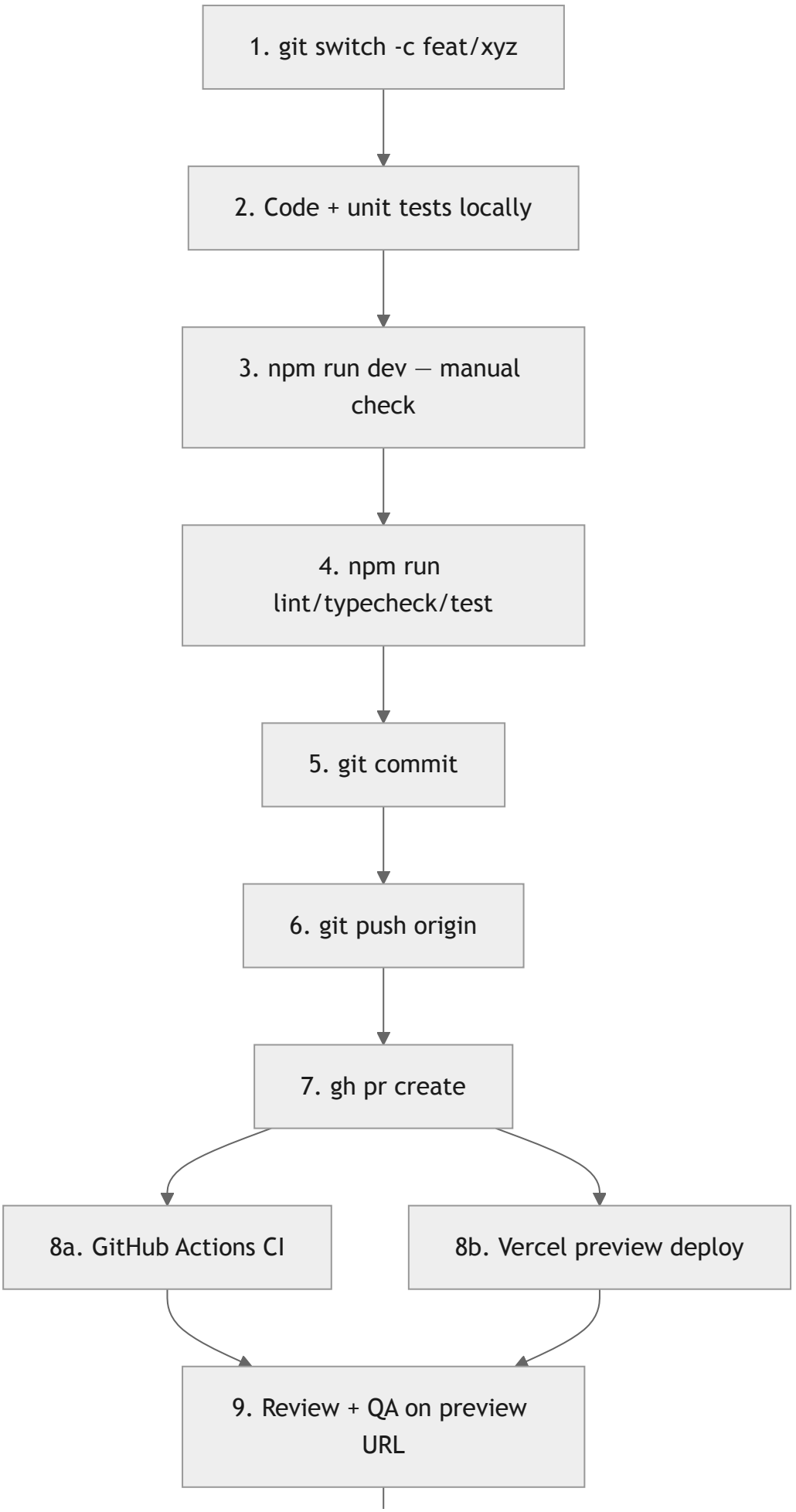
Every table has `owner_id` with an RLS policy scoping reads and writes to `auth.uid()`, except `chat_messages` which inherits ownership via its `thread_id` join (policy: `exists (select 1 from chat_threads t where t.id = thread_id and t.owner_id = auth.uid())`).

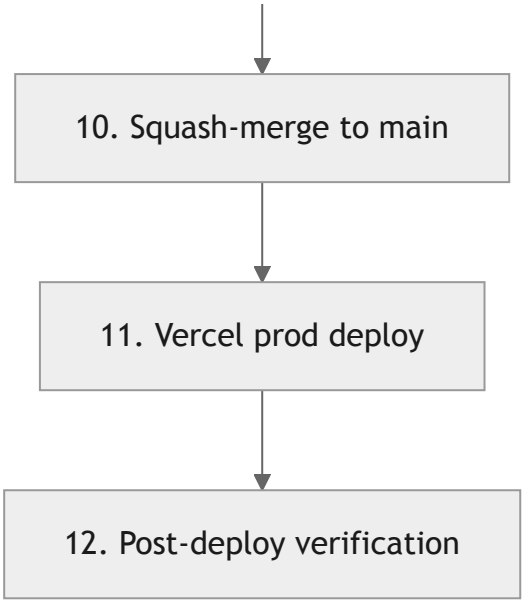
`scenario_versions` gives us cheap history/undo: every `updateScenario(...)` writes a new version row, and `scenarios.current_inputs` is a pointer to the most recent one.

`simulation_runs` lets Monte Carlo results be memoised: re-running with the same `(scenarioId, seed, runs)` returns the cached summary immediately.

How a new feature is shipped

A high-level walkthrough of what happens when you develop a new feature. Twelve steps from idea to live.


1. git switch -c feat/xyz
2. Code + unit tests locally
3. npm run dev — manual check
4. npm run lint/typecheck/test
5. git commit
6. git push origin
7. gh pr create
8a. GitHub Actions CI
8b. Vercel preview deploy
9. Review + QA on preview URL



Step by step

1. Branch off main.

```
git switch main && git pull
git switch -c feat/descriptive-name
```

Short-lived branches only. Nothing lives on a feature branch longer than a few days.

2. Write code + tests locally. Feature-scoped code goes under `app/src/features/<feature>/`. Pure domain logic goes under `packages/core/src/`. Both sides get unit tests (`*.test.ts` / `*.test.tsx`) co-located with the source. New API routes live in `app/src/app/api/<route>/route.ts`.

3. Manual verification against `npm run dev`. Browse to `http://127.0.0.1:3000`. No Supabase / Stripe / Sentry / PostHog keys needed locally — the env schema supplies safe defaults and all external services are dormant.

4. Local gate before pushing.

```
npm run lint
npm run typecheck
npm test
# if the change touches a page or route:
npm run test:e2e
```

Same four commands CI runs. If they pass here, CI will almost always pass too.

5. Commit with a conventional message.

```
git add -A
git commit -m "feat(planner): add Monte Carlo panel


- Wires runMonteCarlo() from @app/core

- Adds histogram chart component

- Behind MONTE_CARLO_ENABLED flag"
```

Prefix with `feat`, `fix`, `chore`, `docs`, or `refactor`.

6. Push the branch.

```
git push -u origin HEAD
```

Upstream is set so subsequent pushes are just `git push`.

7. Open a pull request.

```
gh pr create --fill
```

Or use the GitHub UI. Branch protection rejects direct pushes to `main`.

8a. GitHub Actions runs CI. The `checks` job runs lint, typecheck, unit tests, and a production build (~2 min). The `e2e` job runs the full Playwright suite against a real built Next.js server with browser cache (~1.5 min on cache hit). Defined in `.github/workflows/ci.yml`.

8b. Vercel builds a preview. In parallel with CI, Vercel builds the same commit and posts a unique URL like `https://financial-planner-git-feat-xyz-lejeff.vercel.app`. The URL is pinned to the PR until the branch is deleted. Preview deploys get their own copies of *all* env vars so you can manually test paths that hit Supabase / Stripe / Sentry.

9. Review + QA. A teammate reviews the diff; you (or they) click through the preview URL. If CI is red, fix it. If review asks for changes, push more commits — the preview auto-updates on every push.

> **10. Squash-merge to `main`.** One PR = one commit on `main`. Squash-merge preserves a clean linear history (enforced by branch protection).

> **11. Vercel deploys production.** The squash-merge commit on `main` triggers an automatic production build that replaces `planner.boombaleia.com` when it finishes. Rollback is one click in the Vercel dashboard (previous deployments remain retrievable).

> **12. Post-deploy verification.** Watch the Sentry dashboard for a minute for any new error types. Check PostHog's live events view if you shipped a user-facing change. Smoke-test the core flow manually. Done.

Environments and configuration

6.1 The strict env split

All environment variables flow through two Zod-validated gateways:

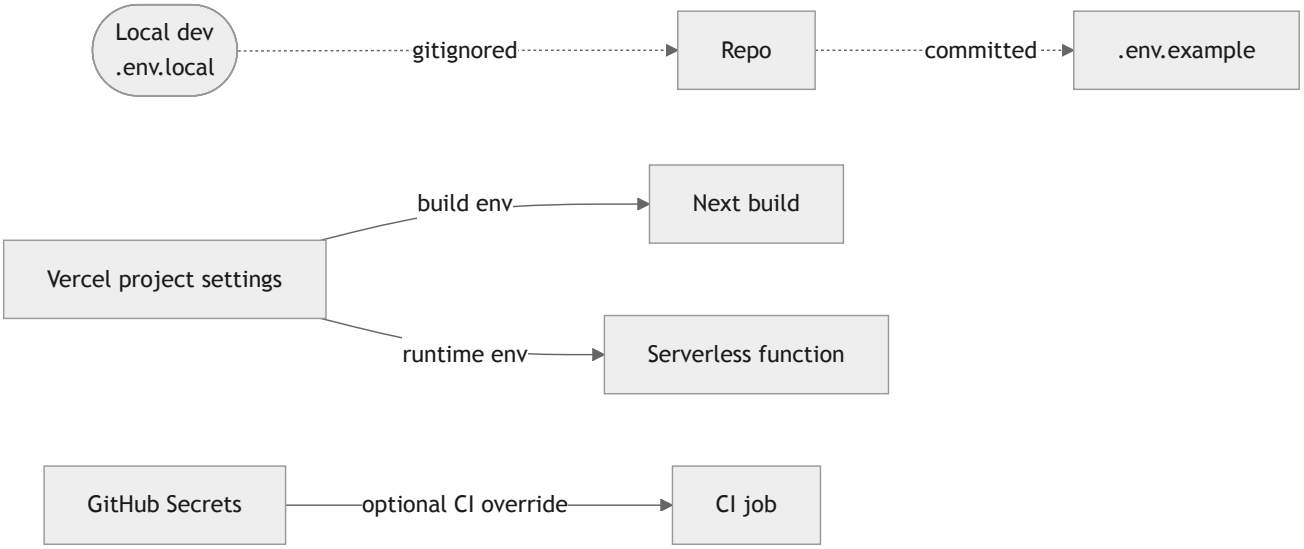
- `app/src/lib/env.ts` — public schema; exposes `NEXT_PUBLIC_*` variables and is safe to import from Client Components.
- `app/src/lib/env.server.ts` — server-only schema, guarded by the `server-only` package. Importing it from a Client Component breaks the Next.js build at compile time.

Both schemas use the same helper:

```
const isDeployBuild = process.env.VERCEL === "1" || process.env.CI === "true";
const optional = (schema, devDefault) => (isDeployBuild ? schema : schema.default(dev
```

Meaning: locally (`npm run dev`, `npm run build` without `CI=true`), safe placeholder defaults are used so you can work without a filled-in `.env.local`. On Vercel and in GitHub Actions, missing values throw at module load and fail the build loudly. This is how we proved the Vercel environment is correctly configured — the first deploy failed on a missing `NEXT_PUBLIC_APP_URL`, and fixing it unblocked the whole pipeline.

6.2 Where secrets live



- Nothing secret is ever committed. `.env.example` is the only source-controlled enumeration of variables.
- Vercel is the authoritative store for production + preview environments.
- GitHub Actions uses well-formed placeholder values in `.github/workflows/ci.yml` — real secrets aren't needed to exercise the build + test paths.

6.3 Per-environment matrix

Scope	Sentry	PostHog	Supabase	Stripe	Notes
Local dev	dormant	dormant	placeholder URL	placeholder key	Everything works without any keys
CI	dormant	dormant	placeholder URL	placeholder key	Strict env schema exercised
Vercel Preview	optional	optional	linked project	test mode	One preview deploy per PR
Vercel Production	on	on	linked project	live mode	Promote via merge to `main`

Security, privacy, data residency

7.1 Data residency

Every stateful hop lives in the EU:

- **Vercel functions** pinned to `fra1` (Frankfurt).
- **Supabase** project in `eu-central-1` (Frankfurt).
- **Sentry** ingest via `ingest.de.sentry.io` (Germany).
- **PostHog** on EU Cloud (`eu.i.posthog.com`, Germany).
- **Resend** sending from the EU region.

Edge-cached HTML is served from the closest POP to the user, but contains no user data — it's either the public landing/planner or a per-user page rendered on demand in `fra1`.

7.2 Tenant isolation

Supabase Postgres RLS is the single source of truth for "who can see what". There is no ad-hoc `where owner_id = $userId` in application code; the session's `auth.uid()` is what the policy checks, and the policy cannot be bypassed by a forgotten filter.

The service-role key (which bypasses RLS) is **server-only**, guarded by `server-only`, and used only for back-office operations like handling Stripe webhooks that need to write without a user session.

7.3 HTTP hardening

Already live on `planner.boombaleia.com` and verified in production:

- `Strict-Transport-Security: max-age=63072000` — 2-year HSTS
- `X-Frame-Options: DENY` — no one can iframe the app
- `X-Content-Type-Options: nosniff` — browsers can't MIME-sniff
- `Referrer-Policy: strict-origin-when-cross-origin`
- TLS is provisioned + renewed by Vercel, HTTP/2 throughout.

Headers come from `app/next.config.ts`'s `headers()` export.

7.4 PII inventory

Service	Contains PII?	Mitigation
Supabase Auth	email	Encryption at rest; EU region; soft-deletable via auth admin
Supabase Postgres	scenarios (may include name)	RLS isolation; per-user export/delete endpoint planned M1
Sentry	stack traces only	Default PII scrubbing on; we never call `Sentry.setUser({ email })`
PostHog	anonymous id only	`persistence: "memory"`, `autocapture: false`, no user identification
Stripe	customer email + payment metadata	PCI offloaded to Stripe; customer portal for user self-service
Resend	email metadata	EU region; 30-day log retention on the free tier
LLM provider	prompts (scenario JSON)	No PII fields sent; opt-out of provider training where available

7.5 Authentication & authorization (M1)

- **AuthN**: Supabase GoTrue email magic-link. HttpOnly cookie session. No long-lived JWT in the client.
- **AuthZ**: Postgres RLS. The app never decides who can see what — the database does.
- **CSRF**: Server Actions are POST-only, same-origin. Next.js adds an internal same-origin check by default in App Router.
- **Rate limiting**: Supabase Auth has built-in rate limiting on sign-in endpoints; we'll add a per-IP limiter in front of `/api/chat` in M3 (and `/api/webhooks/stripe` in M4) using Vercel's KV or Upstash Redis.

Cost model

Free-tier first. Realistic first-paid-tier upgrade points listed below.

Service	Free tier	Upgrade trigger	First paid tier (monthly)
Vercel	Hobby: 100 GB bandwidth, serverless invocations included	> 100 GB bw or > 1M function invocations	Pro: $20
Supabase	Free: 500 MB DB, 50K MAUs, 2 GB bandwidth	> 50K MAUs or paused after 1 week inactivity	Pro: $25
Stripe	Free to create, pay per transaction	Going live with paid subscribers	2.9% + 30¢ per transaction
Resend	Free: 3K emails/month, 100/day	Transactional volume	Pro: $20 (50K emails)
Sentry	Developer: 5K errors + 10K traces	Team with real traffic	Team: $26
PostHog	Free: 1M events, 5K replays	Event volume	Pay as you go: from $0.00005/event
OpenAI	N/A (pay per token)	AI chatbot launches	~ $0.01–$0.03 per chat turn (scales with use)
Gandi (domain)	—	Annual renewal	~ €18/year for `boombaleia.com`

Full-stack M0 cost today: **€0/month + domain renewal**. First paid tier we're likely to cross is Supabase Pro once we're above 50K monthly active users or need Point-in-Time Recovery.

Glossary and further reading

Glossary

- **App Router** — Next.js 13+ routing system based on the `app/` directory. Uses React Server Components by default.
- **Edge** — Vercel's global CDN POPs. Serves static content and runs Edge runtime code close to the user.
- **fra1** — Vercel's region code for Frankfurt, Germany. Where our serverless functions execute.
- **RLS** — Row-Level Security. Postgres feature that attaches a security predicate to every query on a table.
- **Server Action** — a function marked `"use server"` that runs on the server but is called from a client component as if it were local.

- **Tunnel route** — a first-party URL on our domain that proxies to a third-party ingest endpoint to bypass ad blockers (`/monitoring` for Sentry, `/ingest` for PostHog).
- **WIRED-DORMANT** — code path is shipped and correct, but the service stays inert until its credential appears in env.

Further reading

- `README.md` — top-level project intro + scripts
- `infra/m0-deploy-checklist.md` — step-by-step bootstrap of the whole stack
- `infra/vercel-deploy.md` — Vercel setup + custom domain
- `infra/supabase-setup.md` — local CLI + migrations workflow
- `infra/launch-checklist.md` — pre-launch sanity items
- `app/src/lib/env.ts`, `env.server.ts` — env schemas (the compile-time safety net)
- `vercel.json` — region pinning + build config
- `.github/workflows/ci.yml` — CI pipeline definition

Last regenerated: run `npm run docs:build` to rebuild both `docs/architecture.pdf` and `docs/architecture.html` from this markdown source.